

Annotated Bibliography

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The sources below represent some of the main background and data sources I have identified so far, but they are not all of the sources I plan to use for this project. I am still searching for additional datasets that contain information on trout populations, catch rates, fishing effort, and environmental conditions. Because no single dataset appears to contain every variable needed for the project, I plan to merge multiple datasets, likely using location and date as common variables. The Idaho Fish and Game Lake and Streams Survey is currently one of my main potential data sources because it contains fish survey and environmental information that can be downloaded for analysis. The other sources below will primarily provide background information on trout ecology and help determine which environmental variables should be included in my analysis. I am planning on focusing on a specific region rather than the entire US for simplicity.

Harvey, Bret C., Rodney J. Nakamoto, and Jason L. White. "Reduced Streamflow Lowers Dry-Season Growth of Rainbow Trout in a Small Stream." *Transactions of the American Fisheries Society*, vol. 135, 2006, pp. 998–1005. U.S. Forest Service Research and Development, <https://research.fs.usda.gov/treesearch/43928>.

This study investigates how reduced streamflow affects rainbow trout in a northern California stream. The researchers experimentally reduced flow and found substantially lower trout growth in the reduced-flow sections, while also observing differences in the availability of drifting invertebrates. I will use this study as background for why streamflow is an important environmental variable to consider when examining trout conditions and potentially catch rates.

"Lake and Streams Survey." Idaho Fish and Game, <https://idfg.idaho.gov/data/fisheries/lss>. Accessed 15 Sept. 2026.

This will be one of my primary data sources. Idaho Fish and Game's Lake and Streams Survey contains standardized fish survey and limnology data, including information on fish composition, populations, and condition, as well as water quality, temperature/dissolved oxygen profiles, and habitat. The survey data can be downloaded as CSV files. I plan to explore this dataset for trout observations and environmental variables and potentially combine it with other catch, streamflow, or weather datasets as the project develops.

Matthews, Kathleen R., and N. H. Berg. "Rainbow Trout Responses to Water Temperature and Dissolved Oxygen Stress in Two Southern California Stream Pools." *Journal of Fish Biology*, vol. 50, 1997, pp. 50–67. U.S. Forest Service Research and Development, <https://research.fs.usda.gov/treesearch/9765>.

This study examines how rainbow trout respond to high water temperatures and dissolved oxygen levels in two southern California stream pools. The researchers found that trout

distribution was strongly related to water temperature, with fish concentrating in cooler portions of the stream and disappearing from particularly warm habitat. I will use this source to provide ecological background for including variables such as water temperature and dissolved oxygen in my model.

“Robert J. Behnke Papers, 1957–2000.” Archives West, Montana State University Library, <https://archiveswest.orbiscascade.org/ark:80444/xv84500>. Accessed 15 Sept. 2026.

The Robert J. Behnke collection contains research materials related to trout and salmonid ecology, fisheries management, and aquatic environments. The collection includes material on topics such as instream flow, water temperature, dissolved oxygen, aquatic invertebrates, food sources, stream health, and cold-water fisheries. I will use this collection primarily as a background and source-finding resource to better understand the different ecological factors that can influence trout behavior and habitat, and to identify additional research that may help determine which variables should be included in my final analysis.